buildandteach

AUTO LOADER QUICK REFERENCE

Incremental bronze landing

Watch a volume folder. Load only new files into a managed Delta bronze table.
Keep the Auto Loader checkpoint outside the scanned source folder.

1. Schema + two volumes

```
demo_schema = "bt_ingest_" + uuid4().hex[:12]
spark.sql(f"CREATE SCHEMA {schema_fqn}")
spark.sql(f"CREATE VOLUME {schema_fqn}.raw_files")
spark.sql(f"CREATE VOLUME {schema_fqn}.pipeline_state")
# source: /Volumes/.../raw_files/orders/
# checkpoint: /Volumes/.../pipeline_state/.../_checkpoint
```

Actual lesson schema: workspace.bt_ingest_3b5d5d736c66 (your suffix will differ).

2. cloudFiles stream with rescued data

```
(spark.readStream.format("cloudFiles")
  .option("cloudFiles.format","csv").option("header","true")
  .option("rescuedDataColumn","_rescued_data").schema(orders_schema)
  .load(source_dir)
  .select("*", F.col("_metadata.file_path").alias("source_file_path"),
         F.col("_metadata.file_name").alias("source_file"),
         F.current_timestamp().alias("ingested_at_utc"),
         F.lit(run_label).alias("load_run")))
```

Malformed values: typed column null + original text in _rescued_data JSON.

3. Write with availableNow + checkpoint

```
(df.writeStream.option("checkpointLocation", checkpoint_dir)
  .outputMode("append").trigger(availableNow=True)
  .toTable(bronze_fqn))
query.awaitTermination()
```

availableNow is the serverless-friendly trigger. Always awaitTermination.

Runs, idempotency, and verification

Actual Free Edition results

Run 1: 5 rows from files a+b · 0 rescued.

Run 2: only file c · +3 rows · total 8 · no duplicate order_id.

Rescued: order 107 quantity NULL; _rescued_data holds "two" + path.

Run 3: table still 8 · no run3 rows · added 0 (idempotent).

Progress printed 8 on run 3 — that is NOT the new-row count.

SQL PASS: 8 rows / 3 files / 1 rescued / revenue 63.50.

History: v0 CREATE TABLE · v1–v2 STREAMING UPDATE.

4. Prove idempotency with the table

```
run_bronze_ingest("run3_no_new_files")
assert spark.table(bronze_fqn).count() == 8
assert bronze.filter(F.col("load_run")=="run3_no_new_files").count()==0
# Prefer table count over a progress print for "rows added".
```

COPY INTO tracks loaded files in table metadata; Auto Loader uses a checkpoint.

Practice prediction: +2 rows / 2 rescued / revenue 71.00; rerun adds nothing.

Associate foundation: Section 2, August 2026 official guide; partial coverage.
Independent education. No certification guarantee. Fictional-data project.